Volumes by the Disc and Washer Methods

The disc method (about the x-axis)

1 Find the volume of the solid formed by revolving $y = \sqrt{x}$ about the x-axis on $[0, 4]$.

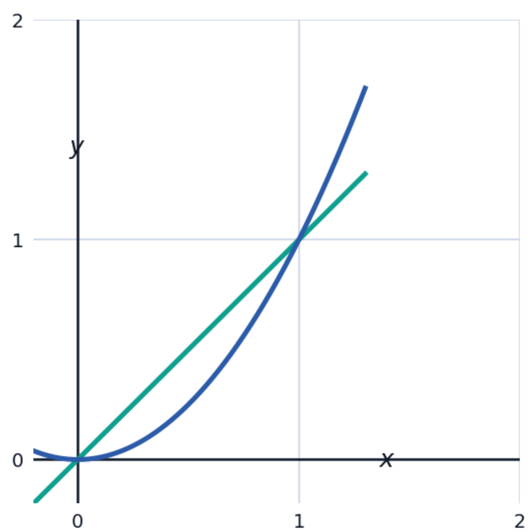
2 Find the volume of the solid formed by revolving $y = x^2$ about the x-axis on $[0, 2]$.

The disc method (about the y-axis)

3 Find the volume formed by revolving $x = y^2$ about the y-axis on $y \in [0, 2]$.

The washer method

4 Find the volume formed by revolving the region between $y = x$ and $y = x^2$ (on $[0, 1]$) about the x-axis.



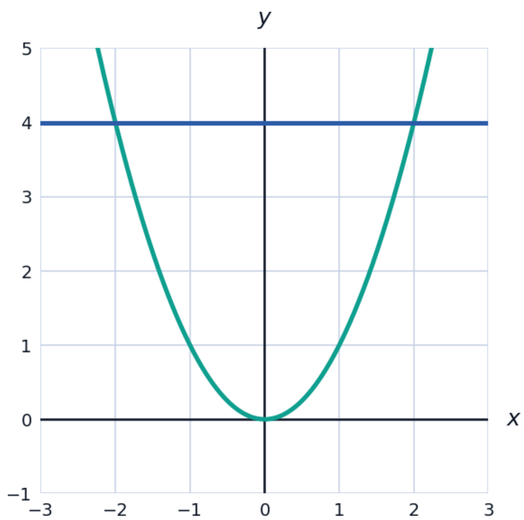
5 Find the volume formed by revolving the region between $y = \sqrt{x}$ and $y = x$ about the x-axis on $[0, 1]$.

Volumes about a horizontal or vertical line (not an axis)

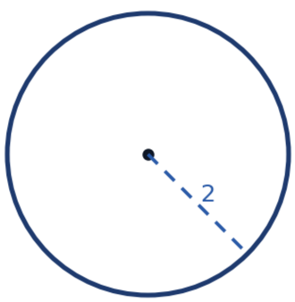
6 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 4$ about the line $y = 4$.

Volumes with known cross-sections

7 The base of a solid is the region between $y = x^2$ and $y = 4$ on $[-2, 2]$. Cross-sections perpendicular to the x-axis are squares. Find the volume.



8 The base of a solid is a circle of radius 2 $(x^2 + y^2 = 4)$. Cross-sections perpendicular to the x-axis are semicircles. Find the volume.



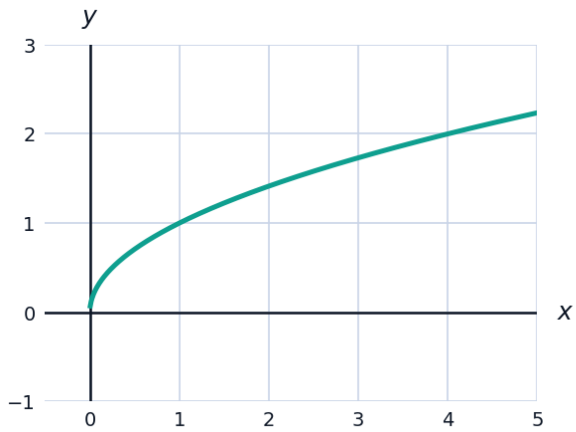
9 Set up (but do not evaluate) the integral for the volume of the solid whose base is the region bounded by $y = \sin x$ and the x-axis on $[0, \pi]$, with cross-sections perpendicular to the x-axis that are equilateral triangles.

Volume by revolving about the y-axis (washer method)

10 Find the volume formed by revolving the region between $x = y^2$ and $x = 2y$ about the y-axis.

Visualizing a solid of revolution's generating region

11 The graph shows $y = \sqrt{x}$ on $[0, 4]$ from the first disc-method problem. Sketching this region mentally revolved about the x-axis produces a solid — use the volume 8π found earlier to find its average cross-sectional area over the interval, and compare that to the cross-sectional area at the midpoint $x = 2$.



A capstone washer-method problem

12 Find the volume formed by revolving the region between $y = 4 - x^2$ and $y = x^2$ about the x-axis (using the region where $4 - x^2 \geq x^2$).

Volumes with rectangular cross-sections

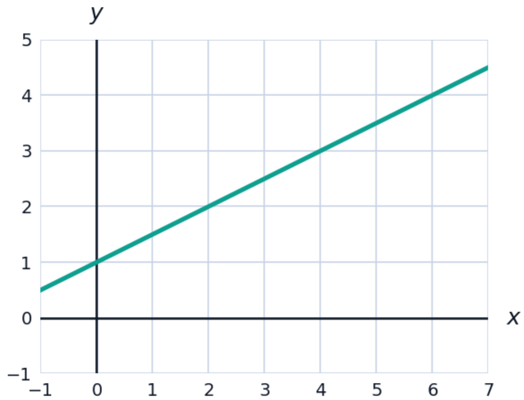
13 The base of a solid is the region bounded by $y = x^2$ and $y = 1$. Cross-sections perpendicular to the x-axis are rectangles with height twice the base. Find the volume.

A capstone volume application

14 A bowl is formed by revolving $y = \frac{x^2}{4}$ (for $0 \leq x \leq 4$, where x is horizontal distance and y is depth from the rim) about the y-axis. Set up (do not evaluate) the disc-method integral for the volume of the bowl in terms of y, using the fact that $x^2 = 4y$.

15 Find the volume formed by revolving $y = e^x$ about the x-axis on $[0, 1]$.

16 A vase is designed by revolving $y = \frac{1}{2}x + 1$ about the x-axis on $[0, 6]$, where x is the height (inches) and y is the radius. Find the volume of the vase.



17 For the region between $y = x$ and $y = x^3$ on $[0, 1]$ (where $y = x \geq y = x^3$), complete each step for the solid formed by revolving this region about the x-axis:

a Identify the outer radius R and inner radius r.

b Set up the washer-method integral.

c Evaluate the integral.